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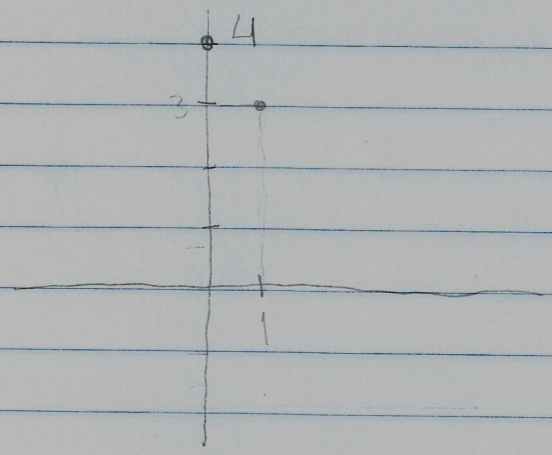
$A = [0,3]$; $B =]1,4]$

1) $A \cap B = [1,3]$

$A \cup B = [0,4]$



2)

a) $|3x+2| = 5$

$3x+2 = 5$ $\quad$ $3x = 5-2 \rightarrow 3x = 3 \rightarrow x = 1$

$3x+2 = -5$ $\quad$ $3x = -5-2 \rightarrow 3x = -7 \rightarrow -\frac{7}{3}$

$x_1 = 1$; $x_2 = -\frac{7}{3}$

b) $|x| = 2x+1$

$x = 2x+1$ $\quad$ $2x+1 = x \rightarrow 1 = x-2x \rightarrow 1 = -x \rightarrow x = -1$

$x = -(2x+1)$ $\quad$ $-1 = x+2x \rightarrow 3x = 1 \rightarrow x = \frac{1}{3}$

$x_1 = \emptyset$; $x_2 = -\frac{1}{3}$

3)

a) $D = \{ x \in \mathbb{R} \mid -4 \leq x \leq 4 \}$
$Im = \{ x \in \mathbb{R} \mid -2 \leq x \leq 3 \}$

b) $D = \{ x \in \mathbb{R} \mid -4 \leq x \leq 3 \}$
$Im = \{ y \in \mathbb{R} \mid 0 < y \leq 4 \}$

c) $f(-4) = -2$ e $g(2) = -2$

d) $f(x)$ é aproximadamente $g(x)$ quando x for -2 ou 2

e) $S = \{ x \in \mathbb{R} \mid 0 < 0 \leq 4 \}$ intervalo descrescente quando $f(x)1 < f(x)2$

f) $S = \{ x \in \mathbb{R} \mid 0 > x \geq -4 \}$ intervalo crescente quando $f(x)1 > f(x)2$

4) $f(x) = |x| - 1$

$0 = |x| - 1$

$|x| = 1$

$x = 1$

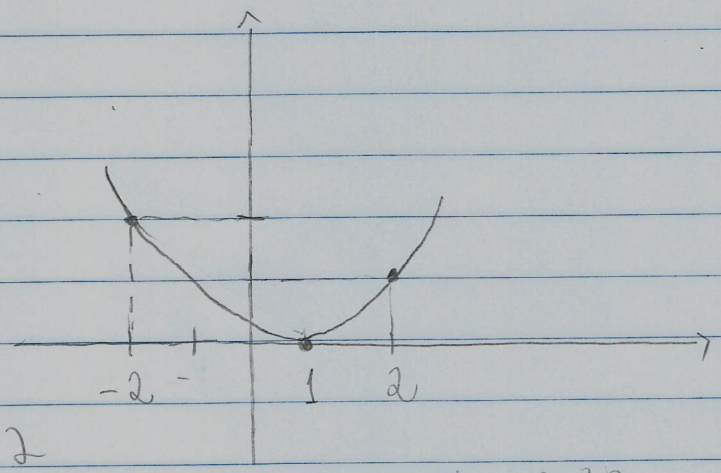
$x = -1$

$\therefore f(x)$ será negativa quando $-1 < x < 1$

$f(x)$ será positiva quando $-1 > x > 1$

$f(x)$ será nula quando x igual a 1 e x igual a -1

5)



se $x \geq 2$

$f(x) = x - 1$

$f(2) = 2 - 1$

$f(2) = 1$

se $0 \leq 2 < 2$

$f(x) = x^2 - 1$

$f(1) = 1^2 - 1$

$f(1) = 1 - 1$

$f(1) = 0$

se $x < 0$

$f(x) = |x|$

$f(-2) = |-2|$

$f(-2) = 2$

módulo